



In Silico Genome-wide Characterization and Abiotic Stress Expression Analysis of the *LEA* Gene Family in *Setaria italica* and *Setaria viridis* Sibling Species

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Abstract

Late embryogenesis abundant (LEA) genes encode a highly diverse family of proteins that play critical roles in responses to biotic and abiotic stresses. Here, we conducted a genome-wide characterization of the *LEA* gene family in the closely related C4 grasses *Setaria italica* and *Setaria viridis*, identifying 106 and 102 *LEA* genes, respectively. These genes were divided into eight *LEA* groups and mapped across the nine chromosomes of both species, with no apparent association between group classification and genomic location. The *LEA2* group emerged as the most abundant, likely due to segmental and tandem duplication events. Most *LEA* genes encode relatively small, highly hydrophilic proteins, although the *LEA2* group display more hydrophobic characteristics and well-defined protein motifs, suggesting functional diversification. Predicted subcellular localization analyses indicated a widespread distribution of *LEA* proteins across cellular compartments, highlighting their potential roles in various cellular compartments. Sequence similarity network and protein conserved motif analyses further underscored the distinct structural signatures within each *LEA* group, supporting their classification and functional differentiation. Metanalyses of RNA-seq data revealed complex and group-specific expression patterns, with *LEA* genes being differentially expressed in *Setaria* spp. under drought, salt, and osmotic stress conditions, suggesting their involvement in key protective and regulatory pathways. Collectively, this genome-wide survey provides a comprehensive resource for understanding the structure, evolution, and stress-responsive functions of *LEA* genes in *Setaria* spp., offering valuable insights for crop breeding enhancing stress tolerance in Poaceae species.

Keywords Late Embryogenesis Abundant Proteins · Poaceae · Drought · Subcellular Localization · Protein Conserved Motifs · RNA-seq

Introduction

Abiotic stresses, such as water deficit and high temperatures, trigger stomatal closure, inhibit photosynthesis, reduce CO₂ assimilation, ultimately disrupting gene expression and impairing plant development and limiting growth and productivity (Osakabe et al. 2014; Peña-Gallardo et al. 2019). Among the genes expressed in response to abiotic stress are those belonging to the Late Embryogenesis Abundant

(*LEA*) family (Bartels 2005; Hong-Bo et al. 2005). *LEA* proteins play a key role in plant defense mechanisms, acting as protectors of biomolecules and membranes under abiotic stress conditions (Xiong and Zhu 2002; Candat et al. 2014; Liang et al. 2016). Based on sequence similarity and functional characteristics, *LEA* proteins were divided into eight groups: *LEA1*, *LEA2*, *LEA3*, *LEA4*, *LEA5*, *LEA6*, Dehydrin, and Seed Maturation Protein (SMP) (Dure 1993; Hundertmark and Hincha 2008; Finn et al. 2009). Most *LEA* proteins are highly hydrophilic and characterized by intrinsically disordered structures (Hong-Bo et al. 2005; Sun et al. 2013). They are predominantly localized in the nucleus, cytoplasm, and chloroplasts of higher plants (Hong-Bo et al. 2005; Candat et al. 2014; Jin et al. 2019). The expression of *LEA* genes was first recorded at the final stage of embryonic development in cotton seeds (Dure et al. 1981),

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and later studies identified the presence of LEA proteins in seeds and vegetative tissues of several other plants (Artur et al. 2019), including *Arabidopsis thaliana* (Hundertmark and Hinch 2008) and the Poaceae *Oryza sativa* (Wang et al. 2007), *Sorghum bicolor* (Nagaraju et al. 2019), *Triticum aestivum* (Bhattacharya et al. 2019), and *Zea mays* (Zhang et al. 2023). These studies showed a significant accumulation of LEA proteins capable of mitigating the damage caused by abiotic stress, emphasizing the importance of these proteins in the genetic improvement of crops important for food and feed.

Setaria italica (foxtail millet) and *Setaria viridis* (green foxtail) are diploid Poaceae ($2n=2x=18$) (Benabdelmouna et al. 2001; Peng and Zhang 2021). *Setaria italica* has been domesticated for more than 11,000 years, probably in China, and has spread throughout Eurasia, Africa and other regions of the world as a source of forage and gluten-free grains with a low glycemic index and rich in micronutrients such as zinc and iron (Yang et al. 2012; Hunt et al. 2021; Muthamilarasan and Prasad 2021; Ereful et al. 2023). On the other hand, *S. viridis*, the wild ancestor of *S. italica*, has been spread as a weed that endanger corn, millet, watermelon and soybean plantations, mainly in northern (Peng and Zhang 2021; Aggarwal et al. 2022). *Setaria italica* and *S. viridis* are increasingly recognized as model C4 Poaceae species due to their favorable attributes for genetic studies, including short life cycles, self-pollination, small genome sizes, and high tolerance to abiotic stress, features that make them more suitable than other C4 photosynthesis Poaceae species such as sorghum and *Panicum* spp. (Doust et al. 2009). Structural and functional characterization studies of LEA genes in *S. italica* and *S. viridis* are scarce. Only one LEA gene (*SiLEA14*, LEA2 group, Seita.3G130700) has been more thoroughly characterized in *S. italica* (Wang et al. 2014). This gene is expressed in several tissues under both normal and abiotic stress conditions, promoting increased tolerance to drought and salt stress when overexpressed in *S. italica*. In addition to this study, transcriptome analyses have identified several LEA genes modulated under abiotic stress conditions in *S. italica*, including Seita.6G114000 (*LEA1*) and Seita.8G115400 (*Dehydrin*) under PEG-induced stress (Qi et al. 2013), and Seita.5G355500 (*LEA2*), Seita.8G039900 (*LEA2*), and Seita.8G115400 (*Dehydrin*) under moderate and severe drought (Suguiyama et al. 2022). In *S. viridis*, several stress-responsive loci have also been reported, such as Sevir.1G011700, Sevir.3G132100.1, and Sevir.3G351000.1, which were shown to be modulated in transcriptomes of plants exposed to heat stress and high light (Anderson et al. 2021), and Sevir.1G270900, Sevir.3G009400, and Sevir.6G002700 under cold and high light stress (Sun et al. 2022). Given the absence of detailed genome-wide of the LEA gene family in *S. italica* and *S.*

viridis, and the importance of these species as model systems for understanding the molecular basis of abiotic stress in C4 Poaceae, this study presents a comprehensive characterization of the physicochemical properties, structural diversity, and gene expression profiles of LEA proteins in *Setaria* species.

Materials and methods

Identification of the LEA Genes in *S. viridis*

The search for LEA sequences in *S. italica* v2.2 and *S. viridis* v1.1, hereafter referred to as *SiLEA* and *SvLEA*, respectively, was conducted via the Phytozome v12 repository (Goodstein et al. 2012) using keywords based on the descriptions of Pfam domains: LEA_1 (PF03760), LEA_2 (PF03168), LEA_3 (PF03242), LEA_4 (PF02987), LEA_5 (PF00477), LEA_6 (PF10714), Dehydrin (PF00257), and SMP (PF04927). Additionally, we searched for LEAs with no Pfam domain annotated using BLASTp available in the Phytozome v12 repository, using LEA sequences from *Brassica napus* (Liang et al. 2016) as queries and an e-value cut-off of $\leq 1E-10$. The presence of Pfam domains was validated using the NCBI CD-Search tool (Wang et al. 2023). To compare the number of LEA genes across forage Poaceae species, we surveyed the genomes of *S. bicolor* v5.1, *Panicum virgatum* v5.1, *Paspalum vaginatum* v3.1, and *Urochloa fusca* v1.1 available in the Phytozome repository, using the same strategy employed for *S. italica* and *S. viridis*.

Characterization of LEA Genes and Proteins

The grand average hydropathy (GRAVY) index, the molecular weight (MW), and the isoelectric point (pI) were determined using ExPASy's ProtParam tool (Artimo et al. 2012). The orthology map was drawn using the PhenoGram v1 tool (Wolfe et al. 2013) and the coordinates of the transcription start site (TSS) of each locus. Schematic diagrams of the gene structure were drawn to scale using the Gene Structure Display Server (Hu et al. 2015). We constructed Sequence Similarity Networks (SSNs) to analyze the relationships among LEA protein sequences. First, we used the EFI-EST FASTA tool (Oberge et al. 2023) to perform an all-by-all BLAST, obtaining similarity scores between sequence pairs and calculating edge values. The SSN network was then visualized using the yFiles Organic Layout tool with a cut-off of $\geq 40\%$ identity, as implemented in the Cytoscape platform (Shannon et al. 2003). The amino acid sequences of *SiLEAs* and *SvLEAs* were analyzed using the Multiple Expectation Maximization for Motif elicitation (MEME) tool (Bailey et al. 2009) to identify the conserved domains

and motifs in each group. The maximum number of motifs was set to 15, with a minimum width of 6 and a maximum width of 50 amino acid residues, and an e -value $\leq 1 \times 10^{-20}$. Protein subcellular localization prediction was conducted using five tools: (i) BUSCA tool (Savojardo et al. 2018) with ‘Eukarya-Plants’ option, BaCelLo (Pierleoni et al. 2006), Plant-mPLOC (Chou and Shen 2010), TargetP, and ChloroP (Emanuelsson et al. 2007). The consensus subcellular localization was determined by compiling the high-confidence predictions from all tools, considering as the most likely the localization site most frequently predicted across the methods employed. The amino acid sequences of LEA2 proteins were aligned using MAFFT v7.505 (Katoh and Standley 2013), as implemented on the CIPRES platform (<https://www.phylo.org/>). Phylogenetic relationships were inferred using the maximum likelihood method with IQ-TREE v2.3.2 (Nguyen et al. 2015), also implemented on the CIPRES platform. The optimal substitution model was evaluated using the ModelFinder tool with the Automatic Model Selection feature, and branch support was evaluated using the SH-like approximate Likelihood ratio test with 1000 replicates.

Gene Expression Profiles

The expression patterns of *S. italica* and *S. viridis* LEA genes under abiotic stress were analyzed by building a database of transcriptomic studies. A literature search was conducted on Google Scholar (<https://scholar.google.com/>) and PubMed (<https://pubmed.ncbi.nlm.nih.gov/>) using the

keywords [(*Setaria italica*) OR (Foxtail millet) OR (*Setaria viridis*)] and [(stress) OR (tolerance)] and [(transcriptome) OR (transcriptomics) OR (expression) OR (RNA-Seq)]. Previously published studies with available RNA sequencing Log₂FoldChange (LogFC) data for *S. italica* and *S. viridis* were compiled into an expression value database. Loci were defined as differentially expressed if they met an False Discovery Rate-adjusted (FDR-adjusted) P-value of ≤ 0.05 and a LogFC cutoff of ≤ -1.0 or ≥ 1.0 criteria. Additionally, we generated heatmaps for tissues under non-stress conditions using Fragments Per Kilobase of transcript per Million mapped reads (FPKM) values from control transcriptome libraries. Heatmaps were generated using the ComplexHeatmap package (Gu et al. 2016) in R Studio v4.3.1 (Posit team 2025), with the Euclidean method for clustering and complete linkage for distance calculation.

Results

LEA Loci in *Setaria italica* and *Setaria viridis* Display Distinct Diversity Patterns Across Groups

The LEA loci of *S. italica* and *S. viridis* were identified using a combination of BLAST and keyword search methods. This approach allowed the identification of 106 loci in *S. italica* and 102 loci in *S. viridis*, resulting in a total of 208 loci divided into the eight LEA groups based on sequence homology and the presence of conserved Pfam protein motifs (Supplementary Table 1). Loci were sequentially named based on their chromosomal locations. The majority of loci belong to the LEA2 group, with 130 genes, followed by the SMP, LEA3, Dehydrin, LEA1, LEA4, LEA5, and LEA6 groups, which contain 18, 17, 14, 10, 10, 7, and 2 genes, respectively (Table 1). The number of loci and the distribution of LEA genes among groups in *S. italica* and *S. viridis* are similar to those observed in other species of the Panicoideae subfamily (Fig. 1). Among the species analyzed, *P. virgatum* had the highest number of LEA loci (188), while *U. fusca* had the lowest (90).

The gene length, protein length, and molecular weight (MW) of the SiLEA and SvLEA proteins in the LEA2 and LEA4 groups are greater than those in the other groups (Supplementary Table 1). MW ranged from 7.69 to 56.55 kDa, with smaller peptides for SiLEA37 (LEA3) and SvLEA43 (LEA1), and larger peptides for SiLEA66 (LEA2) and SvLEA63 (LEA2). The isoelectric point (pI) revealed that 75.48% of the proteins have a basic nature ($pI > 7$), with a mean pI of 9.39 ± 0.83 , predominantly found in the LEA1, LEA2, LEA3, LEA4, and Dehydrin groups (Fig. 2 and Supplementary Table 1). The remaining 24.52% of the proteins, primarily from the LEA5, LEA6 and SMP groups, exhibit

Table 1 General features of the *S. italica* (Si) and *S. viridis* (Sv) LEA loci

Group	Number of genes	Mean $\pm$ SD			Number of introns (Min-Max)
		Gene length (bp)	Protein length (aas)	Molecular weight (kDa)	
<i>SiDehydrin</i>	7	1137 $\pm$ 306	214 $\pm$ 73	21.85 $\pm$ 8.03	1–3
<i>SvDehydrin</i>	7	1078 $\pm$ 320	211 $\pm$ 74	21.60 $\pm$ 8.14	1–2
<i>SiLEA1</i>	5	619 $\pm$ 134	128 $\pm$ 31	13.18 $\pm$ 3.12	0–1
<i>SvLEA1</i>	5	1379 $\pm$ 1380	128 $\pm$ 31	13.17 $\pm$ 3.13	0–4
<i>SiLEA2</i>	65	1464 $\pm$ 954	241 $\pm$ 58	25.98 $\pm$ 6.26	0–3
<i>SvLEA2</i>	65	1469 $\pm$ 928	240 $\pm$ 54	25.86 $\pm$ 5.78	0–3
<i>SiLEA3</i>	9	615 $\pm$ 334	98 $\pm$ 15	10.37 $\pm$ 1.84	0–1
<i>SvLEA3</i>	8	723 $\pm$ 270	116 $\pm$ 44	12.23 $\pm$ 4.73	0–1
<i>SiLEA4</i>	5	1493 $\pm$ 124	319 $\pm$ 88	33.23 $\pm$ 10.49	1–3
<i>SvLEA4</i>	5	1563 $\pm$ 126	319 $\pm$ 88	33.22 $\pm$ 10.48	1–2
<i>SiLEA5</i>	4	548 $\pm$ 283	123 $\pm$ 30	13.18 $\pm$ 3.74	0–1
<i>SvLEA5</i>	3	849 $\pm$ 160	114 $\pm$ 28	12.10 $\pm$ 3.75	0–1
<i>SiLEA6</i>	1	353	117	12.32	0
<i>SvLEA6</i>	1	353	117	12.32	0
<i>SiSMP</i>	10	854 $\pm$ 375	207 $\pm$ 52	20.66 $\pm$ 5.50	0–2
<i>SvSMP</i>	8	1203 $\pm$ 281	230 $\pm$ 50	22.97 $\pm$ 5.34	1–2

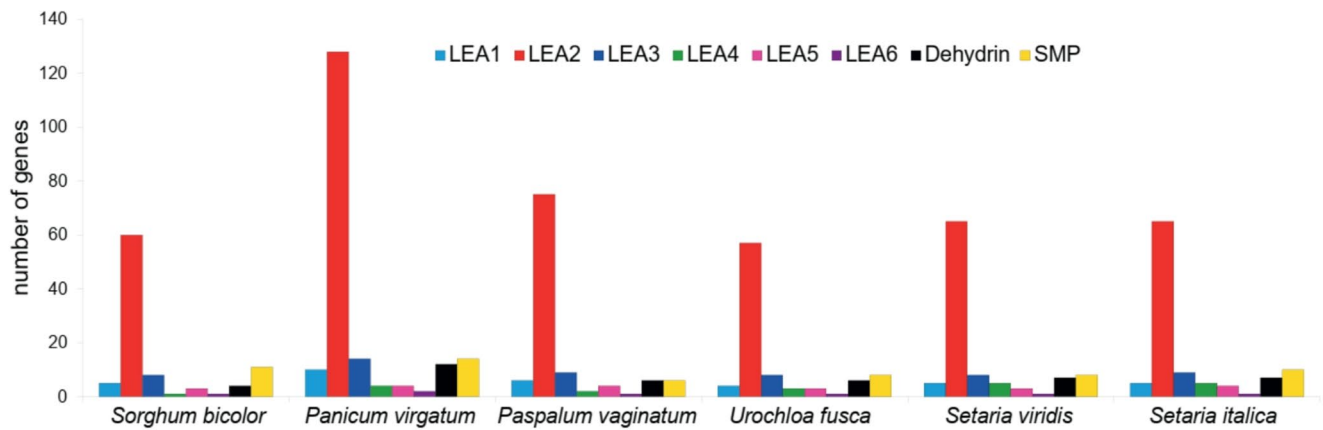
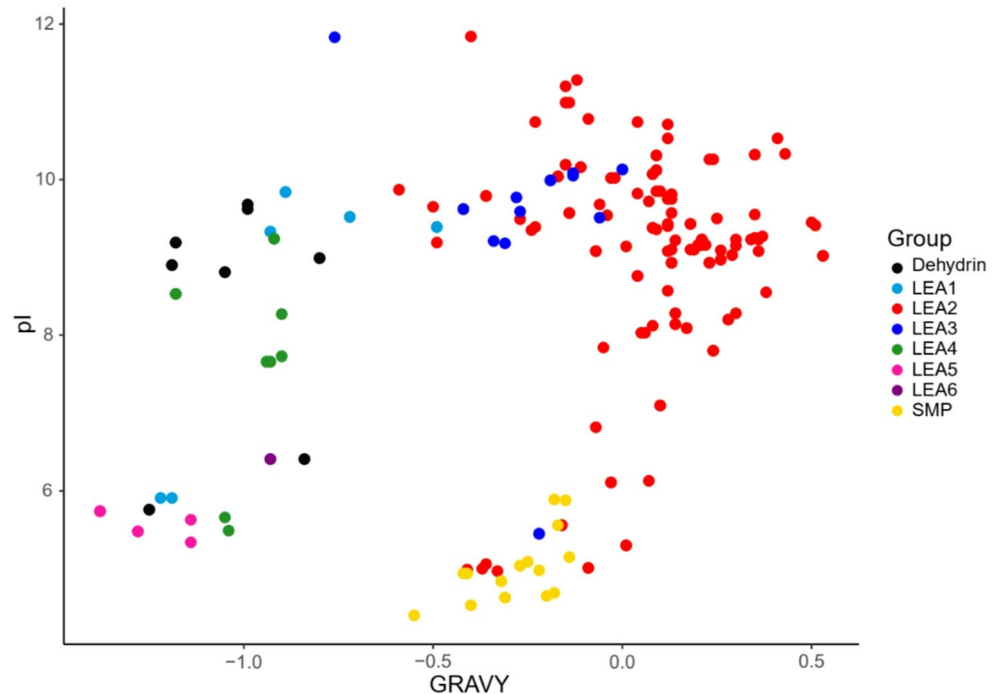


Fig. 1 Comparison of the number of *LEA* genes among *S. bicolor*, *P. virgatum*, *P. vaginatum*, *U. fusca*, *S. italica*, and *S. viridis*

Fig. 2 Distribution of pI and GRAVY values for all *LEA* proteins identified in *S. italica* and *S. viridis*. Each circle represents a predicted protein, with colors indicating the different *LEA* groups



an acidic nature ($pI < 7$), with a mean pI of 5.43 ± 0.57 . Except for LEA2 proteins, LEA proteins exhibited negative GRAVY index scores, indicating that the SiLEA and SvLEA groups are primarily hydrophilic. In contrast, 88 out of 130 members of the LEA2 group showed positive GRAVY scores, suggesting a predominance of hydrophobicity.

The SiLEA and SvLEA genes were distributed across all nine chromosomes in both species (Fig. 3). Chromosome 5 had the highest number of genes, with 19 SiLEA and 20 SvLEA genes, while chromosome 4 had the fewest, with five SvLEA and four SiLEA genes. The genome distribution analysis also revealed several tandem duplications, with 34% and 27% of the genes duplicated in tandem in *S. italica* and *S. viridis*, respectively, showing a greater predominance of the LEA2 group. Tandem duplications were also observed

in the SMP, LEA5, and Dehydrin groups. Although most LEA genes had orthologous counterparts between *S. italica* and *S. viridis*, we identified some species-specific genes and duplications. For example, *S. italica* had a specific tandem duplication of SMP genes on chromosome 1 involving four genes. Additionally, we identified an inversion on chromosome 4 involving members of the SMP and LEA1 groups.

The structural diversity of SiLEA and SvLEA genes was analyzed by comparing their gene structures, revealing a high level of conservation between orthologous genes in *S. italica* and *S. viridis*. Over half of the LEA genes (60%) have no introns, with 103 members belonging to the LEA2 group (Supplementary Fig. 1). Of the 208 SiLEA and SvLEA genes, 66 contain only one exon, 15 contain two exons and two contain three exons. Most of these genes have both 5' and 3'

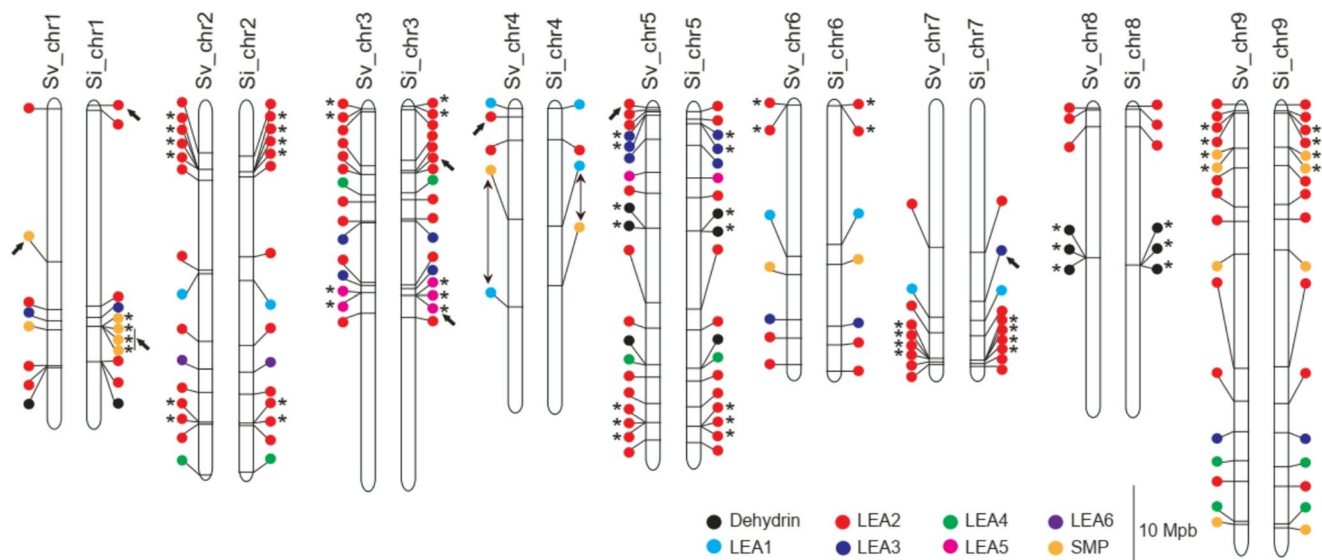


Fig. 3 Chromosomal location analysis of the *S. italica* and *S. viridis* LEA genes. The physical mapping of the genes was based on data retrieved from Phytozome. Chromosomes were drawn to scale and chromosome numbers were indicated above each chromosome ideogram. Genes belonging to different groups are indicated by different

untranslated regions (UTRs). Analysis of Pfam domains in SiLEA and SvLEA proteins revealed similar patterns within groups, though variations in domain size and quantity were observed among groups (Supplementary Fig. 2). The number and size of Pfam domains were highly conserved in orthologous pairs, with most proteins containing a single Pfam domain specific to each LEA group, ranging from 78 to 525 amino acids. Only members of the LEA4, Dehydrin, and SMP groups displayed repeated domains.

LEA Groups Represent a Diverse Evolutionary Lineage

Understanding the evolutionary relationship of the LEA proteins is a challenging task. Different LEA groups possess distinct protein domains, which indicates the low similarity among the sequences and complicates the generation of nucleotide and amino acid alignments that could clarify the phylogenetic relationships between and within groups. Consequently, phylogenetic trees of LEA proteins generally lack branch support values (Jin et al. 2019; Zan et al. 2020; Jia et al. 2023; Zhang et al. 2023), making it difficult to propose robust phylogenetic relationships. In this work, we have decided to analyze the clustering patterns between LEA sequences using a SSN instead of reconstructing a phylogenetic tree. The SSN was based on the results of all-by-all BLAST searches and sequences with no or very low (<40%) BLAST similarity were not grouped in the same network. The SSN analysis organized SiLEA and SvLEA protein sequences into multiple independent networks, with

colors, which follow the same color pattern used for Fig. 1. Single black arrows indicate genes for which an orthologous pair could not be identified. Double black arrows indicate gene location inversions. Tandemly duplicated genes are marked with asterisks

no instances of sequences from different groups occupying the same network (Fig. 4). In some groups, particularly LEA1, LEA2, and LEA4, multiple networks were observed, suggesting low similarity between sequences within the same group. Conversely, the LEA3, LEA5, and SMP groups formed networks that included the majority of sequences, indicating a higher level of similarity among them.

To expand the understanding of sequence diversity within LEA groups, we analyzed the presence of protein sequence motifs within each group individually. The number of conserved motifs ranged from 3 to 15 per group (Fig. 5). The LEA2, SMP, LEA4, and Dehydrin groups exhibited the highest numbers of conserved motifs, while the LEA6 group had the fewest. Interestingly, we observed a high variability of motifs between sequences of some groups, aligning with the high number of networks observed in the previous analysis (Fig. 4). The LEA2 group alone formed at least 12 subgroups with distinct motif patterns. One notable subgroup, comprising SvLEA11, SvLEA12, SiLEA14, SiLEA15, SiLEA16, SvLEA13, SiLEA92, SvLEA88, and SiLEA93 sequences, contains a unique long conserved motif absent in other members of group (motif 5). This motif is located in the C-terminal region of the LEA_2 superfamily domain (PF03168).

LEA Proteins Function across Various Subcellular Compartments

The subcellular localization of LEA proteins was analyzed using five distinct in silico prediction tools, and a consensus



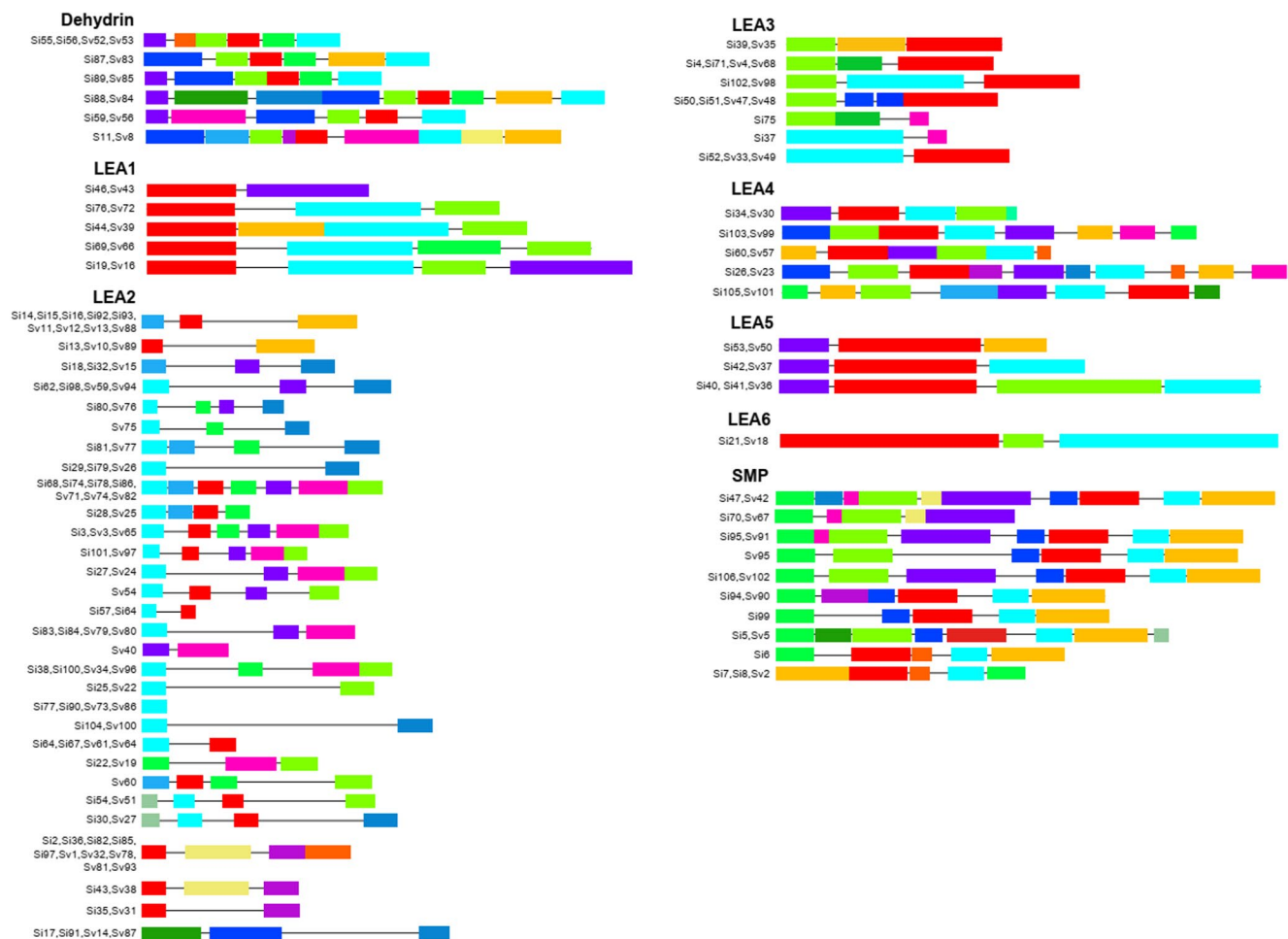


Fig. 5 Conserved motif analysis of LEA proteins. The black lines represent non-motif regions, and the colored bars represent conserved motifs in each group. Sequences were classified based on the presence of identical conserved motifs. Schematic representations were drawn to scale

(Fig. 7). We did not observe a complete correlation between the phylogenetic relationships and predicted subcellular localizations. The phylogenetic tree divided the sequences into six main clades. Clades A and B, the most basal groups, were composed of sequences predominantly predicted to localize to the nucleus. Clade C included two sequences with predicted targeting to the plasma membrane and one sequence to the endomembrane system. Sequences in Clade D were predicted to localize both to nucleus and chloroplast. In contrast, Clades E and F comprised sequences with predicted localizations to multiple subcellular compartments, primarily the nucleus and chloroplast, suggesting a potential diversification in functional targeting within these groups.

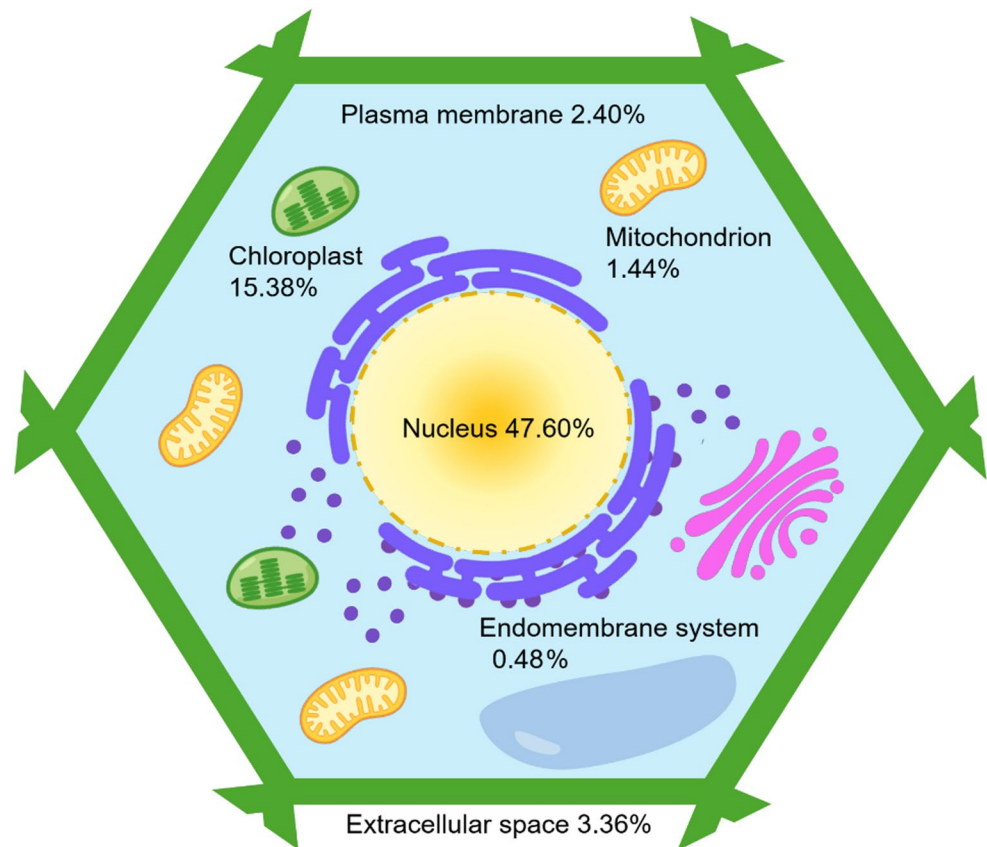
LEA Genes Exhibit Specific Responsiveness To a Variety of Abiotic Stressors and Tissues

To investigate the expression profiles of *LEA* genes under abiotic stress, a meta-analysis was conducted using transcriptome data from previously published studies in *S.*

italica and *S. viridis*. Literature searches were carried out between October 2020 and September 2024 via Google Scholar (<https://scholar.google.com/>) and PubMed (<https://pubmed.ncbi.nlm.nih.gov/>) (Supplementary Table 2). LogFC values were used to generate a qualitative overview of gene expression differences between experimental and control conditions (hereafter referred to as “samples”). A total of 17 studies providing LogFC data were retrieved, enabling the analysis of 50 samples (Supplementary Table 2). For *S. italica*, the dataset included 22 samples under water stress, 10 under salt stress, three under potassium deficiency, two under PEG-induced osmotic stress, and one under nitrogen deficiency. For *S. viridis*, the dataset comprised six samples under cold stress and three samples each under heat and high light stress.

The results allowed us to identify clusters of *LEA* genes responsive to abiotic stress. A cluster of 21 genes was found to be upregulated under water stress, salt stress, and PEG-induced osmotic stress in shoots, roots, and seeds in *S. italica* (Fig. 8A). These genes include representatives from several

Fig. 6 In silico prediction of the subcellular localization of *S. italica* and *S. viridis* LEA proteins. Schematic representation of a general plant cell highlighting the various subcellular compartments where LEA proteins were predicted to be located



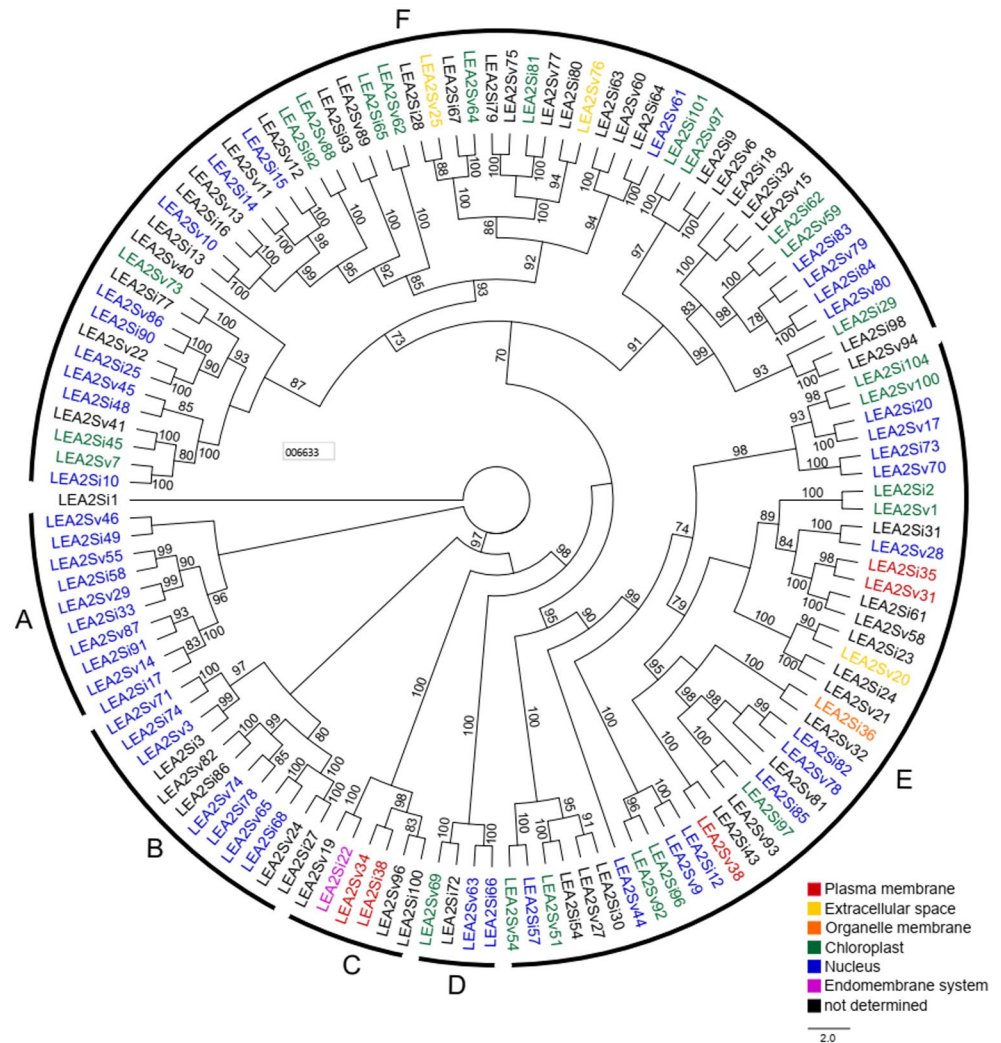
LEA groups: five from each of the *Dehydrin* (Seita.1G267200, Seita.5G166200, Seita.5G166300, Seita.8G115200, and Seita.8G115400), *LEA1* (Seita.2G159600, Seita.4G007900, Seita.4G140300, Seita.6G114000, and Seita.7G217700), and *LEA4* (Seita.2G437900, Seita.3G162400, Seita.5G287800, Seita.9G420500, and Seita.9G526800) groups; four *SMP* (Seita.1G212300, Seita.4G159500, Seita.6G123500, and Seita.9G533500) genes; and two *LEA2* (Seita.3G269000, Seita.5G014300) genes (Supplementary Table 2). Interestingly, there was a lack of differential expression and downregulation of *LEA2*, *SMP* and *LEA3* genes across various abiotic stress conditions and tissues analyzed. A cluster of seven *LEA2* genes was identified (Seita.3G008300, Seita.3G015800, Seita.5G309600, Seita.5G355500, Seita.5G392200, Seita.7G296300, and Seita.7G296400), exhibiting downregulation in roots under drought conditions (Supplementary Table 2). In *S. viridis*, differentially expressed genes were predominantly from the *LEA2* and *LEA3* groups, with high-light and cold stress having the most significant impact on gene expression patterns (Fig. 8B). Additionally, we performed a qualitative analysis of the expression patterns of *LEA* genes in control groups, using FPKM values available from previously published studies (Supplementary Table 2). A total of 16 RNA-seq libraries from *S. italica* and one from *S. viridis* were included, comprising seven root, four leaf, four stem, and one seedling

libraries. Roots and leaves libraries grouped separately (Fig. 8C). Roots exhibited the highest number of transcriptionally active *LEA* genes, with a predominance of *LEA2*, *Dehydrin*, *LEA3*, and *LEA4* groups (Fig. 8C–D).

Discussion

LEA proteins are associated with plant growth, development, and responses to various stresses (Hong-Bo et al. 2005). Although members of the *LEA* gene family have been identified in many plant species, a genome-wide analysis of these genes in *Setaria* spp. has not yet been conducted. In this study, we identified 106 and 102 *LEA* genes in *S. italica* and *S. viridis*, respectively, distributed across eight *LEA* groups. The proportion of *S. italica* and *S. viridis* *LEA*-encoding genes in each group aligns with the distribution reported in other Panicoideae species (Fig. 1), as well as in several eudicot species such as *Gossypium hirsutum* (Magwanga et al. 2018), *Camellia sinensis* (Jin et al. 2019), *Solanum lycopersicum* (Jia et al. 2022), and *Glycine max* (Guo et al. 2023). A high degree of synteny was observed between the *LEA* genes of *S. italica* and *S. viridis* (Fig. 3). Only four *LEA2*, four *SMP*, one *LEA3*, and one *LEA5* genes lacked orthologous counterparts. *LEA* genes showed similar exon and intron architecture within the groups, being common

Fig. 7 Phylogenetic analysis of the *LEA2* gene group. The phylogenetic tree was reconstructed using the IQ-TREE tool, employing the maximum likelihood method with the WAG+F+R4 substitution model. The analysis was based on a 710 aa alignment. Letters denote the primary clades referenced in the text, and SH-like test support values ($\geq 70\%$) are indicated at each node. Si: *S. italica*, Sv: *S. viridis*. Colors represent the consensus subcellular localization predictions



a low number or absence of introns (Supplementary Fig. 1), as previously reported for other plant species (Jia et al. 2023; Li et al. 2023; Lin et al. 2024). Rapid responses under stressful conditions may be associated with genes with few introns because reducing introns facilitates protein synthesis and reduces energy consumption (Lan et al. 2013; Cheng et al. 2021; Jia et al. 2022; Zhang et al. 2023).

LEA proteins from different groups contain distinct Pfam domains (Supplementary Table 1), which poses challenges to performing a comprehensive multiple sequence alignment and, consequently, to reconstruct a robust phylogeny that could reliably infer evolutionary relationships among LEA groups. To better understand patterns of sequence divergence within and between LEA groups, we constructed a BLAST-based SSN (Fig. 4). The SSN revealed a clear separation of LEA groups, with no network containing sequences from more than one group. Due to this high level of sequence divergence, each LEA group in *S. italica* and *S. viridis* was analyzed separately to highlight group-specific structural and functional characteristics. Whereas SSN

clustering and protein motif conservation provided insights into the sequence diversity of LEA proteins, the *in silico* prediction of subcellular localization and gene expression analyses can contribute to the understanding of the functional roles of each LEA group. *In silico* analyses rely on computational models and require cautious interpretation, as they only indicate the most likely subcellular targets and gene expression patterns.

Dehydrins are the most extensively studied group of LEA proteins and have been associated with multiple protective functions for various biomacromolecules, including membrane lipids, proteins, and DNA, due to their intrinsically disordered nature and structural versatility (Malik et al. 2017; Smith and Graether 2022). In *Setaria* species, Dehydrins were predominantly hydrophilic and basic (Fig. 2), and were predicted to localize mainly to the nucleus by the *in silico* analysis. This is consistent with other *in silico* and functional studies (Hernández-Sánchez et al. 2015). The SSN grouped all Dehydrins into a single network, except for the orthologous pairs SiLEA059/SvLEA056 and

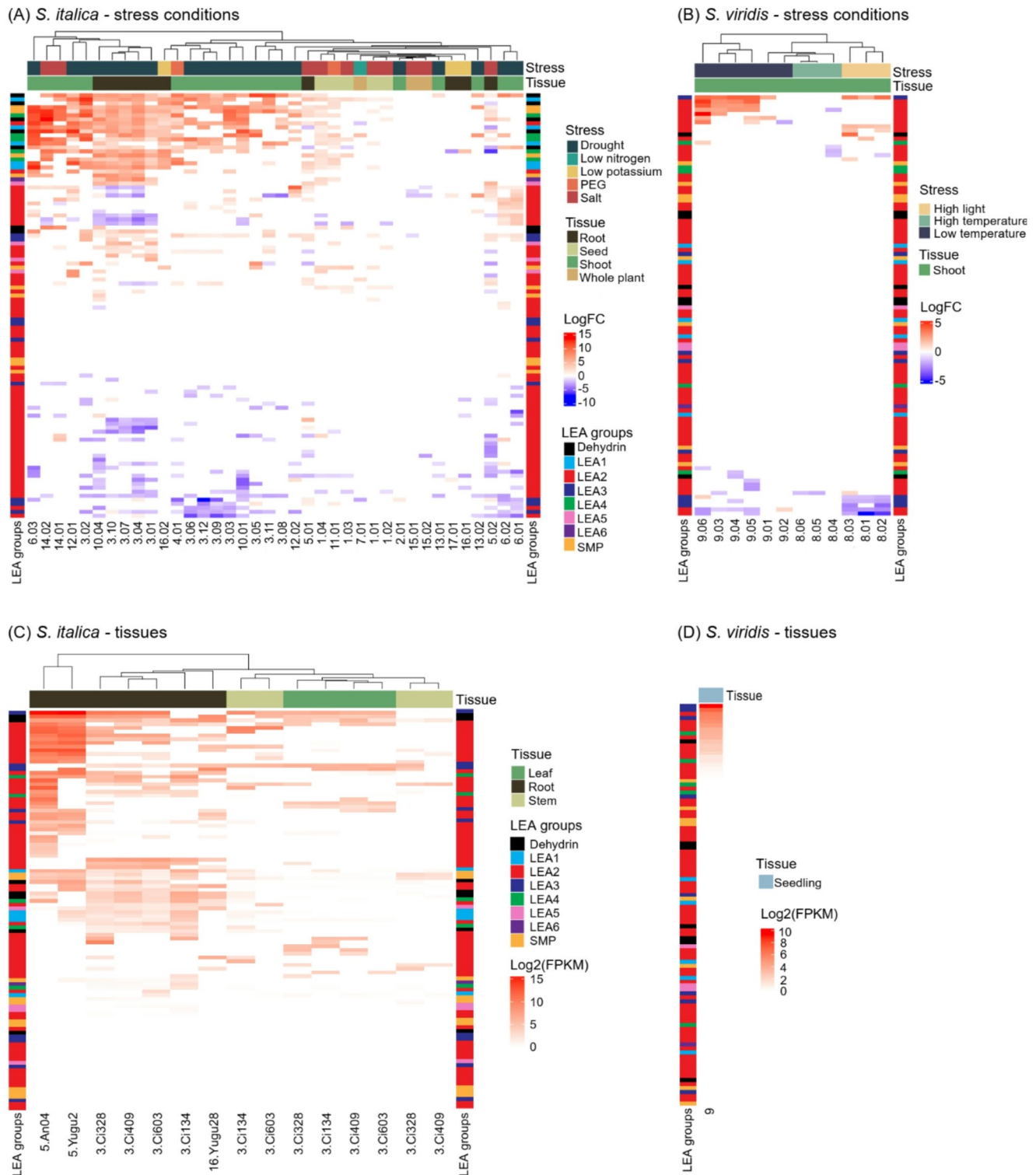


Fig. 8 Heatmaps of *LEA* gene expression: (A) LogFC values under stress conditions in *S. italica*; (B) LogFC values under stress conditions in *S. viridis*; (C) FPKM values across tissues of *S. italica*; and (D) FPKM values across tissues of *S. viridis*. Color bars on the left/right and above the heatmaps show the *LEA* groups and stress conditions, respectively. Each number below the heatmaps corresponds to a com-

parison between a stress condition and its respective control (A–B), and to transcriptome library from tissues under non-stress condition (C–D) (Supplementary Table 2). The data was clustered for both genes and comparisons; however, only the dendrogram for sample comparisons is displayed

SiLEA011/SvLEA008 (Fig. 4), a pattern further supported by the conserved protein domain analysis (Fig. 5). Dehydrins have been shown to be positively regulated under drought and cold stress in wheat and barley (Kosová et al. 2014), in leaves of *S. italica* under drought (Suguiyama et al. 2022), in maize kernels (Zhang et al. 2023), and in rice under ABA treatment, drought, osmotic, salt, and cold stress (Wang et al. 2007). Our gene expression analysis further supports the role of *Dehydrins* under drought and PEG stress in the C4 drought-tolerant species *S. italica* and *S. viridis*, although no upregulation was observed in response to low-temperature stress.

LEA5, LEA6, and SMP groups shared several features with Dehydrins, including their clustering into single SSN networks (Fig. 4), low motif divergence (Fig. 5), hydrophilic and basic nature (Fig. 2), frequent upregulation under osmotic stress, and nuclear subcellular localization (Supplementary Table 1). *S. viridis* and *S. italica* contain a higher number of *LEA5* genes compared with rice, sorghum, and maize (Wang et al. 2007; Nagaraju et al. 2019; Zhang et al. 2023). LEA6, in turn, is the most recently evolved LEA group, restricted to angiosperms (Artur et al. 2019) and absent from rice and maize (Wang et al. 2007; Zhang et al. 2023). The presence of LEA6 and the expanded LEA5 family in *Setaria* may contribute to the enhanced tolerance of these species to drought and salt stress (Doust et al. 2009). By contrast, SMPs appear to be the oldest LEA group (Artur et al. 2019), acting as general players in the abiotic stress response across Poaceae (Huang et al. 2018; Nagaraju et al. 2019; Zan et al. 2020; Zhang et al. 2023). Collectively, these findings, together with previous reports, support a protective role for LEA5, LEA6, and SMPs, along with Dehydrins, in plant dehydration response (Pantelić et al. 2022).

LEA1 and LEA4 sequences were also characterized by high hydrophilicity (Fig. 2), predominant nuclear localization predictions (Supplementary Table 1), and frequent upregulation under osmotic stress (Fig. 8A). However, they exhibited a much more pronounced sequence diversification pattern, as revealed by both SSN (Fig. 4) and conserved motif analyses (Fig. 5). It remains unclear whether this high level of sequence divergence translates into functional diversity, although previous studies have demonstrated roles of *LEA1* and *LEA4* genes in osmotic stress tolerance. For instance, silencing of a *LEA1* gene reduced drought tolerance in *A. thaliana* (Olvera-Carrillo et al. 2010), while the *ZmLEA3* gene (LEA4 group) was reported to protect proteins from osmotic and oxidative stress (Liu et al. 2013) and to stabilize membranes and proteins under low-temperature conditions (Liu et al. 2016).

Among the LEA3 proteins with predicted subcellular localizations, all were associated with either mitochondria or chloroplasts (Supplementary Table 1). The SSN clustered

most LEA3 sequences into a single network, except for two *S. italica* and two *S. viridis* proteins (Fig. 4). These more divergent members lacked an N-terminal region (Fig. 5), were predicted to localize to mitochondria, and were upregulated under abiotic stress (Fig. 8A–B). In contrast, most of the other *LEA3* genes in the main SSN cluster were either downregulated (*SiLEA004*, *SiLEA050*, *SiLEA051*, *SiLEA102*, *SvLEA047*, *SvLEA048*, and *SvLEA098*) or not differentially expressed under abiotic stress conditions. Consistent with our findings, previous studies reported upregulation of *LEA3* genes in maize and sorghum in response to low temperature, drought, and other osmotic stresses (Liu et al. 2016; Nagaraju et al. 2019). Altogether, these results indicate that *LEA3* genes may contribute to abiotic stress tolerance in Poaceae by functioning in energy metabolism organelles, while also exhibiting gene-specific patterns of subcellular localization and transcription that require further experimental validation.

LEA2 proteins are considered atypical, as they comprise the most abundant and hydrophobic LEA group, with more defined secondary structures in solution compared to other LEA families (Artur et al. 2019). LEA2 have been classified into distinct sequence communities (Artur et al. 2019), which is consistent with the results of our phylogenetic analysis (Fig. 7) as well as the highly structured profiles observed in the SSN and conserved protein motif analyses (Figs. 4 and 5). This high representation is likely due to segmental and tandem duplications, key mechanisms driving the expansion of the *LEA* gene family (Liang et al. 2016; Artur et al. 2019; Xu et al. 2020). Most of the LEA2 proteins in clades A and B were predicted to localize to the nucleus (Fig. 7). In contrast, LEA2 proteins from other clades were predicted to occur in multiple compartments, mainly the nucleus and chloroplast, but also in membranes and the extracellular space, consistent with previous reports in tomato, wheat, and banana (Bhattacharya et al. 2019; Piyatissa and Bandupriya 2021; Jia et al. 2022). However, this diversity was not directly associated with abiotic stress responses, as most *LEA2* genes were downregulated or not differentially expressed under osmotic, light, and temperature stress (Fig. 8A–B), although several genes were transcriptionally active under non-stress conditions (Fig. 8C–D). Altogether, our results suggest that *LEA2* proteins are not strongly modulated by abiotic stress in *Setaria* spp., but may play roles across several cellular compartments and tissues through hydrophobic interactions and molecular associations within plant cells.

Conclusion

In this study, we identified and characterized *LEA* genes and proteins from *S. italica* and *S. viridis*, two closely related C4 perennial Poaceae species. The SiLEA and SvLEA proteins

were divided into the eight canonical LEA groups, displaying considerable diversity in gene structure and protein properties both within and between groups. This diversity suggests significant variability in protein activity, subcellular localization, and an evolutionary history toward functional diversification. Gene expression analysis revealed that *SiLEA* and *SvLEA* genes are broadly responsive to abiotic stresses, as has been observed in several other plant species. However, our results demonstrate that individual members within each LEA group exhibit distinct structural features and gene expression profiles, which should be considered to better understand the mechanisms underlying abiotic stress tolerance in C4 Poaceae species. These findings provide a valuable foundation for future efforts to improve abiotic stress tolerance in the Poaceae species, particularly regarding water deficit, high salinity, and extreme temperature conditions.

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Author Contributions GOS, LBS, CS and NS analyzed the data, prepared figures, tables, and approved the final draft. GOS and NS authored and reviewed article drafts. NS conceived and designed the workflow of analysis data.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Conflict of interest The authors declare no conflict of interest.

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